

Case Study 14

Pavlov's Conditioned Response Experiments

Russian scientist Ivan Petrovich Pavlov was born in 1849. Although he won a Nobel Prize for his studies of the digestive system, Pavlov is often best known for his “drooling dog” experiments. When a dog encounters food, saliva starts to pour from the salivary glands located in the back of the oral cavity. Saliva is needed to make food easier to swallow and it also contains enzymes that break down certain compounds such as starch.

Pavlov became interested in studying reflexes when he noticed that the dogs in his lab drooled in the absence of food -- the logical **stimulus** for drooling. Pavlov discovered the dogs were reacting to lab coats. Each time the dogs were fed, the person who served the food was wearing a lab coat. In essence, the dogs reacted (drooled) whenever they saw a lab coat because they associated the coat with being fed.

Reflexes

A **reflex** is a response to a stimulus. For example, when light strikes our eyes, our pupils shrink or contract. In situations where there is little light, our pupils expand or dilate. This type of reflex or response is **unconditioned**. Unconditioned reflexes are natural or built-in, and the body responds in the same fashion every time the stimuli (the light for example), is changed. Drooling by dogs in the presence of food is an unconditioned reflex or response. In a series of experiments, Pavlov tried to determine how drooling could be conditioned by unnatural stimuli (such as lab coats).

	In one of his experiments Pavlov struck a bell when the dogs were fed. If the bell was sounded in close association with their meals, the dogs learned to associate the sound of the bell with food. After repeated trials of the bell and food being presented together, the dogs would salivate when the bell was sounded in the absence of food. These experiments were first conducted in the 1890s and 1900s.
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The Results

Pavlov's discovered that environmental events (bell) that previously had no relation to a given reflex (drooling) could, through experience, trigger the reflex (drooling). This form of learned response is called a conditioned reflex. The process whereby dogs (or other animals) learn to connect a stimulus to a reflex is called **conditioning**. Food aversion is an example of a natural conditioned reflex. For example, if an animal eats something with a distinctive taste and then experiences nausea and vomits, the animal will not be eager to eat food with the same distinctive taste. Linking nausea to taste is an evolutionarily successful strategy.

As a result of his experiments, Pavlov proved that animals (including humans) could be trained to respond in a certain way to a particular stimulus. An equally important principle in conditioned learning is that an established conditioned response (salivating in the case of the dogs) decreases in intensity if the conditioned stimulus (bell) is repeatedly presented without the unconditioned stimulus (food). This process is called **extinction**.

Conditioning forms the basis for much learned human behaviour. Commercial advertising, for example, effectively associates a product's name (stimuli) to a given response (positive emotion). In one taste test, elementary school children were given two identical hamburgers. One hamburger was in a plain wrapper and one was in a McDonald's wrapper. The vast majority of children indicated that the burger in the McDonald's wrapper tasted better. The response in this case (satisfaction) may have been influenced by the McDonald's wrapper (stimuli) although the burgers were identical. Do you think fast food companies want to condition this type of response?